

TASK S5.01

CONSULTAS MONGODB

Descripció

Treballarem amb una base de dades que conté col·leccions relacionades amb una aplicació d'entreteniment cinematogràfic:

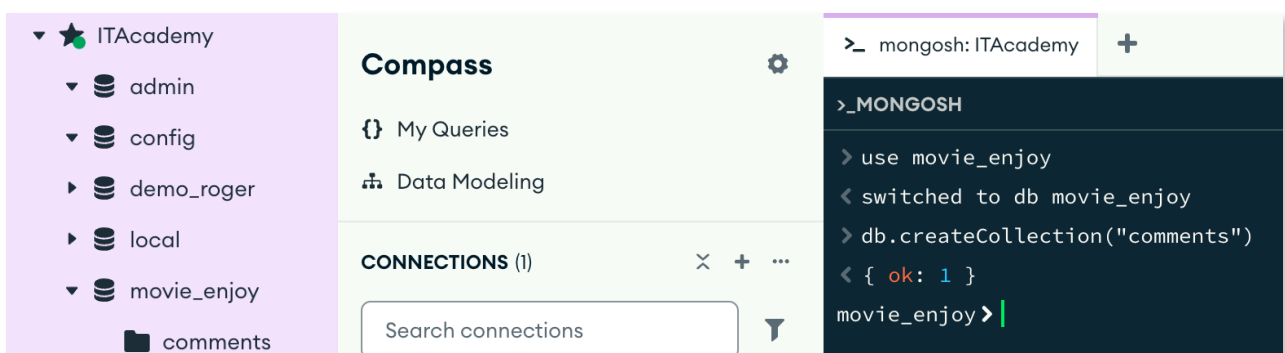
- users: Emmagatzema informació d'usuaris/es, incloent-hi noms, emails i contrasenyes xifrades.
- theatres: Conté dades de cinemes, com ID, ubicació (direcció i coordenades geogràfiques).
- sessions: Guarda sessions d'usuari, incloent-hi ID d'usuari i tokens JWT per a l'autenticació.
- movies: Inclou detalls de pel·lícules, com a trama, gèneres, durada, elenc, comentaris, any de llançament, directors, classificació i premis.
- comments: Emmagatzema comentaris d'usuaris/es sobre pel·lícules, amb informació de l'autor/a del comentari, ID de la pel·lícula, text del comentari i la data.

Duràs a terme algunes consultes que et demana el client/a, el qual està mesurant si seràs capaç o no de fer-te càrrec de la part analítica del projecte vinculat amb la seva base de dades.

Level 1

Crea una base de dades amb MongoDB utilitzant com a col·leccions els arxius adjunts.

Level 1 - Database creation



```
rogerdefez@MacBook-Air-de-Roger ~ % mongoimport --db movie_enjoy --collection comments --file '/Users/rogerdefez/Documents/Cursos i Llibres/BootCamp IT Academy/03_Especialitzacio/Sprint_5_DB_No_Relacionals/Dades originals_S5/comments.json'
2026-03-05T20:33:48.558+0100 connected to: mongod://localhost/
2026-03-05T20:33:49.278+0100 50304 document(s) imported successfully, 0 document(s) failed to import.
rogerdefez@MacBook-Air-de-Roger ~ % mongoimport --db movie_enjoy --file '/Users/rogerdefez/Documents/Cursos i Llibres/BootCamp IT Academy/03_Especialitzacio/Sprint_5_DB_No_Relacionals/Dades originals_S5/movies.json'
2026-03-05T20:49:18.125+0100 no collection specified
2026-03-05T20:49:18.127+0100 using filename 'movies' as collection
2026-03-05T20:49:18.136+0100 connected to: mongod://localhost/
2026-03-05T20:49:19.095+0100 23539 document(s) imported successfully, 0 document(s) failed to import.
rogerdefez@MacBook-Air-de-Roger ~ % mongoimport --db movie_enjoy --file '/Users/rogerdefez/Documents/Cursos i Llibres/BootCamp IT Academy/03_Especialitzacio/Sprint_5_DB_No_Relacionals/Dades originals_S5/sessions.json'
2026-03-05T20:50:36.430+0100 no collection specified
2026-03-05T20:50:36.431+0100 using filename 'sessions' as collection
2026-03-05T20:50:36.438+0100 connected to: mongod://localhost/
2026-03-05T20:50:36.462+0100 1 document(s) imported successfully, 0 document(s) failed to import.
rogerdefez@MacBook-Air-de-Roger ~ % mongoimport --db movie_enjoy --file '/Users/rogerdefez/Documents/Cursos i Llibres/BootCamp IT Academy/03_Especialitzacio/Sprint_5_DB_No_Relacionals/Dades originals_S5/theaters.json'
2026-03-05T20:52:05.709+0100 no collection specified
2026-03-05T20:52:05.710+0100 using filename 'theaters' as collection
2026-03-05T20:52:05.716+0100 connected to: mongod://localhost/
2026-03-05T20:52:05.732+0100 1564 document(s) imported successfully, 0 document(s) failed to import.
rogerdefez@MacBook-Air-de-Roger ~ % mongoimport --db movie_enjoy --file '/Users/rogerdefez/Documents/Cursos i Llibres/BootCamp IT Academy/03_Especialitzacio/Sprint_5_DB_No_Relacionals/Dades originals_S5/users.json'
2026-03-05T20:52:56.529+0100 no collection specified
2026-03-05T20:52:56.530+0100 using filename 'users' as collection
2026-03-05T20:52:56.539+0100 connected to: mongod://localhost/
2026-03-05T20:52:56.567+0100 185 document(s) imported successfully, 0 document(s) failed to import.
rogerdefez@MacBook-Air-de-Roger ~ %
```

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

ITAcademy

- admin
- config
- demo_roger
- local
- movie_enjoy
 - comments
 - movies
 - sessions
 - theaters
 - users

ITAcademy > movie_enjoy > comments

Documents 0 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

25 1 - 25 of 50304

```
{
  "_id": ObjectId('5a9427648b0beeb69579d3'),
  "name": "Cameron Duran",
  "email": "cameron_duran@fakegmail.com",
  "movie_id": ObjectId('573a1390f29313caab4217'),
  "text": "Quasi dicta culpa asperiores quaeat perferendis neque. Est animi pari..."
  "date": 1983-04-27T20:39:15.000+00:00
}
```

```
{
  "_id": ObjectId('5a9427648b0beeb69579ea'),
  "name": "Cameron Duran",
  "email": "cameron_duran@fakegmail.com",
  "movie_id": ObjectId('573a1390f29313caab4d33d'),
  "text": "Ad asperiores mollitia aperiam non incidunt. Totam fugiat cumque praes..."
  "date": 1989-01-21T14:20:47.000+00:00
}
```

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My Queries

Data Modeling

CONNECTIONS (1)

Search connections

ITAcademy

- admin
- config
- demo_roger
- local
- movie_enjoy
 - comments
 - movies
 - sessions
 - theaters
 - users
- sample_airbnb
- sample_analytics
- sample_geospatial
- sample_guides

ITAcademy > movie_enjoy > movies

Documents 24K Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

25 1 - 25 of 23539

```
{
  "_id": ObjectId('573a1390f29313caab4d323'),
  "plot": "A young boy, oppressed by his mother, goes on an outing in the country ...",
  "genres": Array (3),
  "runtime": 14,
  "rated": "UNRATED",
  "cast": Array (4),
  "num_mflix_comments": 2,
  "poster": "https://m.media-amazon.com/images/M/MV5BMTMzMDCxMjgyNl5BMl5BanBnXkFtZT...",
  "title": "The Land Beyond the Sunset",
  "fullplot": "Thanks to the Fresh Air Fund, a slum child escapes his drunken mother ...",
  "languages": Array (1),
  "released": 1912-10-28T00:00:00.000+00:00,
  "directors": Array (1),
  "writers": Array (1),
  "awards": Object,
  "lastupdated": "2015-08-29 00:27:45.437000000",
  "year": 1912,
  "imdb": Object,
  "countries": Array (1),
  "type": "movie",
  "tomatoes": Object
}
```

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My Queries
Data Modeling

CONNECTIONS (1)

Search connections

- ITAcademy
 - admin
 - config
 - demo_roger
 - local
 - movie_enjoy
 - comments
 - movies
 - sessions**
 - theaters
 - users

ITAcademy > movie_enjoy > sessions

Documents 1 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#) [Explain](#) [Reset](#) [Find](#) [Options](#)

25 1 - 1 of 1

```
{ "_id": ObjectId('5a97f9c91c807bb9c6eb5fb4'),  
  "user_id": "t3qulfeem@kwiv5.6ur"  
}
```

```
{  
  "jwt": "eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpYXQiOiJlMTk5MDkzMjEsIm5iZiI6Im..."  
}
```

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My Queries
Data Modeling

CONNECTIONS (1)

Search connections

- ITAcademy
 - admin
 - config
 - demo_roger
 - local
 - movie_enjoy
 - comments
 - movies
 - sessions
 - theaters**
 - users

ITAcademy > movie_enjoy > theaters

Documents 1.6K Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#) [Explain](#) [Reset](#) [Find](#) [Options](#)

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```
{ "_id": ObjectId('59a47286cfa9a3a73e51e72e'),  
  "theaterId": 1008  
}
```

```
{ "_id": ObjectId('59a47286cfa9a3a73e51e736'),  
  "theaterId": 1017  
}
```

```
{ "_id": ObjectId('59a47286cfa9a3a73e51e730'),  
  "theaterId": 1002  
}
```

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My Queries
Data Modeling

CONNECTIONS (1)

Search connections

- ITAcademy
 - admin
 - config
 - demo_roger
 - local
 - movie_enjoy
 - comments
 - movies
 - sessions
 - theaters
 - users**

ITAcademy > movie_enjoy > users

Documents 185 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#) [Explain](#) [Reset](#) [Find](#) [Options](#)

25 1 - 25 of 185

```
{ "_id": ObjectId('59b99db9cfa9a34dcd7885bf'),  
  "name": "Jon Snow",  
  "email": "kit_harington@gameofthron.es",  
  "password": "$2b$12$fDEu1Ru66tLWAVidMN.b0.929BlfnyqdGuhWMyzf0Af/ATY0yLoY6"
```

```
{ "_id": ObjectId('59b99db9cfa9a34dcd7885c0'),  
  "name": "Sansa Stark",  
  "email": "sophie_turner@gameofthron.es",  
  "password": "$2b$12$ncIVE81..AtA0ysPZkL19.G5V0EdIwvsZh1f18lxWEr3d1pG/Uusi"
```

```
{ "_id": ObjectId('59b99db9cfa9a34dcd7885c1'),  
  "name": "Arya Stark",  
  "email": "maisie_williams@gameofthron.es",  
  "password": "$2b$12$19h3KjeTg3/sPNFHztDb6eGwKrcIwLVXVsh9POS15PS9kS1guZEq"
```

Level 1 - Database creation and initialization

Para la importación del Json comments he usado el comando incluyendo 'collection' debido a que ya tengo creada la colección en mi DB de mongoDB

```
mongoimport --db movie_enjoy --collection comments --file "comments.json"
```

Para movies.json y el resto de Json, uso la ventaja que Mongoimport crea la colección en la DB en caso de que no se le indique --collection

```
mongoimport --db movie_enjoy --file "movies.json"
```

Step 1 - Database creation and initialization

Character definition for enhanced compatibility with “ñ” and other characters in MySQL 8.

```

8  -- Database creation and initialization
9  • CREATE DATABASE starmarketplace
10 CHARACTER SET utf8mb4; -- Character Definition for Enhanced Compatibility with "ñ" and Other Characters in MySQL 8
11
12 • USE starmarketplace
13 ;

```

	Time	Action	Response
✓ 1	01:00:01	CREATE DATABASE starmarketplace CHARACTER SET utf8mb4	1 row(s) affected
✓ 2	01:00:01	USE starmarketplace	0 row(s) affected

Step 2 - Table creation:

As a general rule I will create all the fields of all the tables with DataType VARCHAR with this I will avoid errors in the data loading. Later I will adjust the DataType to its real data type. The PK fields, mainly 'id' if its possible I will put them as INT and the other numerical fields I will put it INT, DECIMAL or FLOAT as appropriate or DATE or TIMESTAMP if is date. I also would like to comment that I incorporate the clause “IF NOT EXIST” as a good praxis to avoid any error in case that the table already exists.

‘products’ table - dimension table:

- This table allows us to create the 'id' as INT
- Important to highlight the 'price' field we must load as VARCHAR since the \$ symbol makes it impossible to load as a DECIMAL field

```

17  -- Creation table 'products'
18  • CREATE TABLE IF NOT EXISTS products (
19      id INT,
20      product_name VARCHAR(255),
21      price VARCHAR(255),
22      colour VARCHAR(50),
23      weight VARCHAR(50),
24      warehouse_id VARCHAR(50)
25  );

```

	Time	Action	Response
✓ 1	21:29:15	CREATE TABLE IF NOT EXISTS pr...	0 row(s) affected